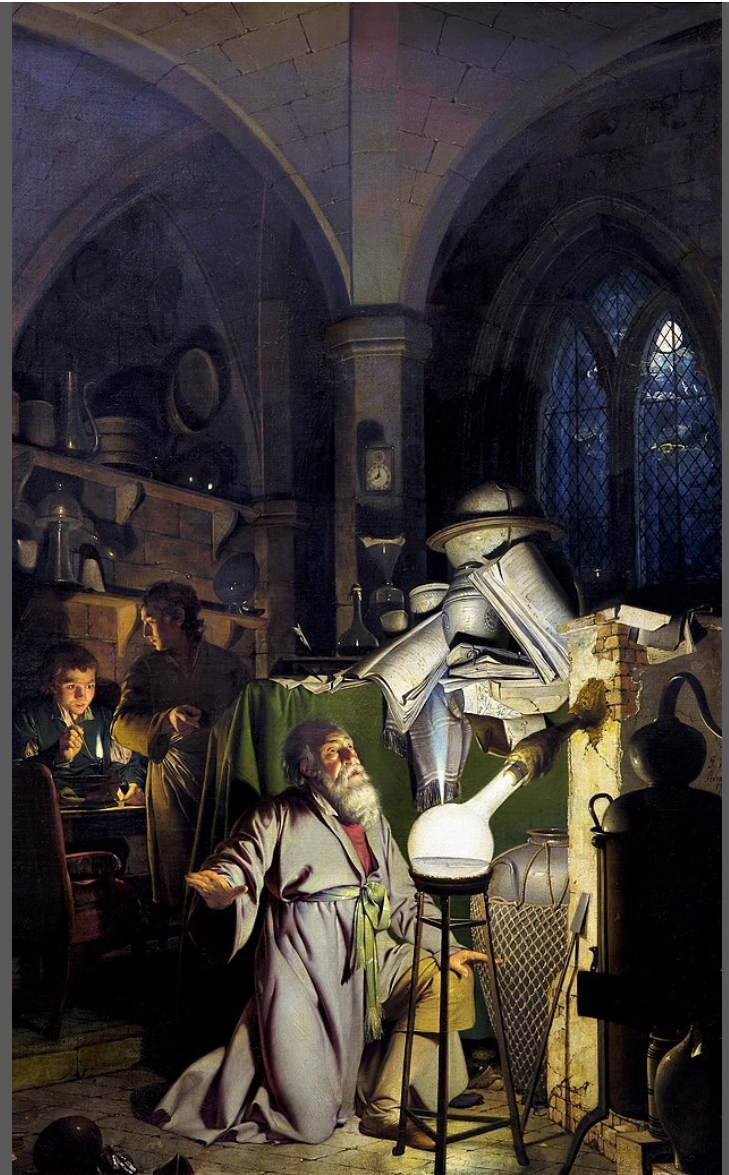


Jordens alder II

Fagets videnskabsteori og -etik: Fysik og geoscience
Aarhus Universitet

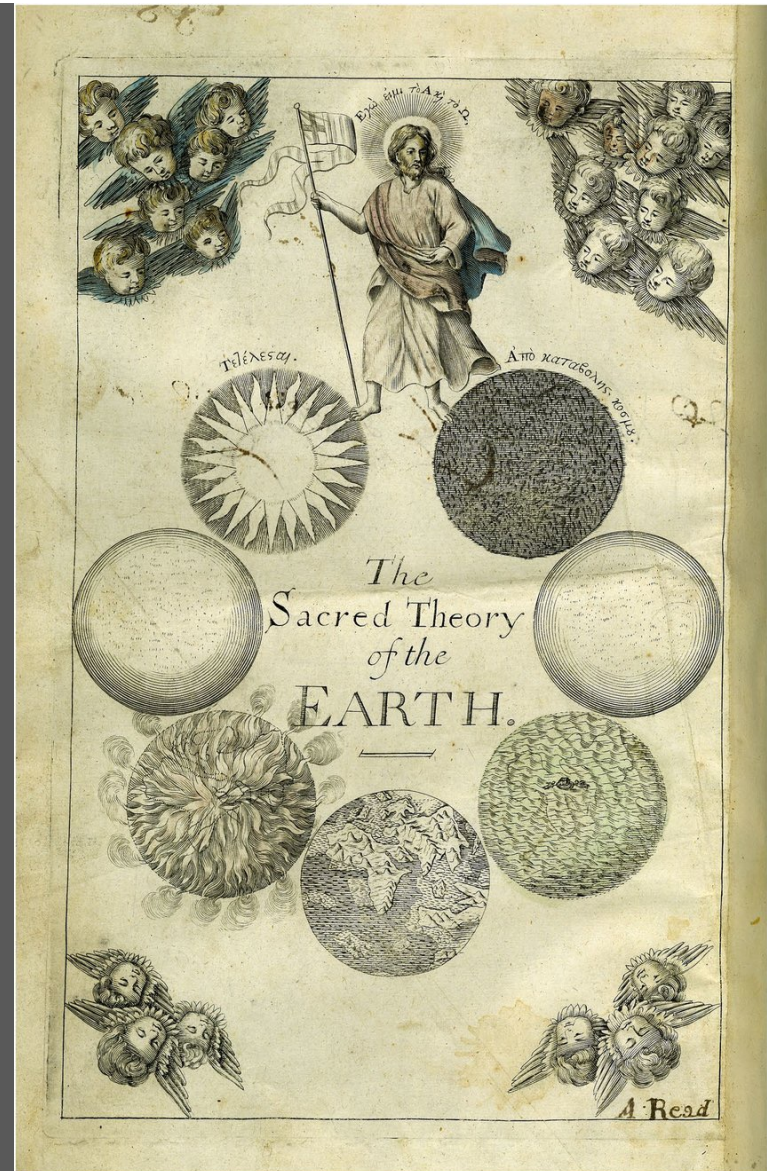
Torsdag den 29. januar 2026

Lektor Christoffer Basse Eriksen
Center for Videnskabsstudier
cbe@css.au.dk



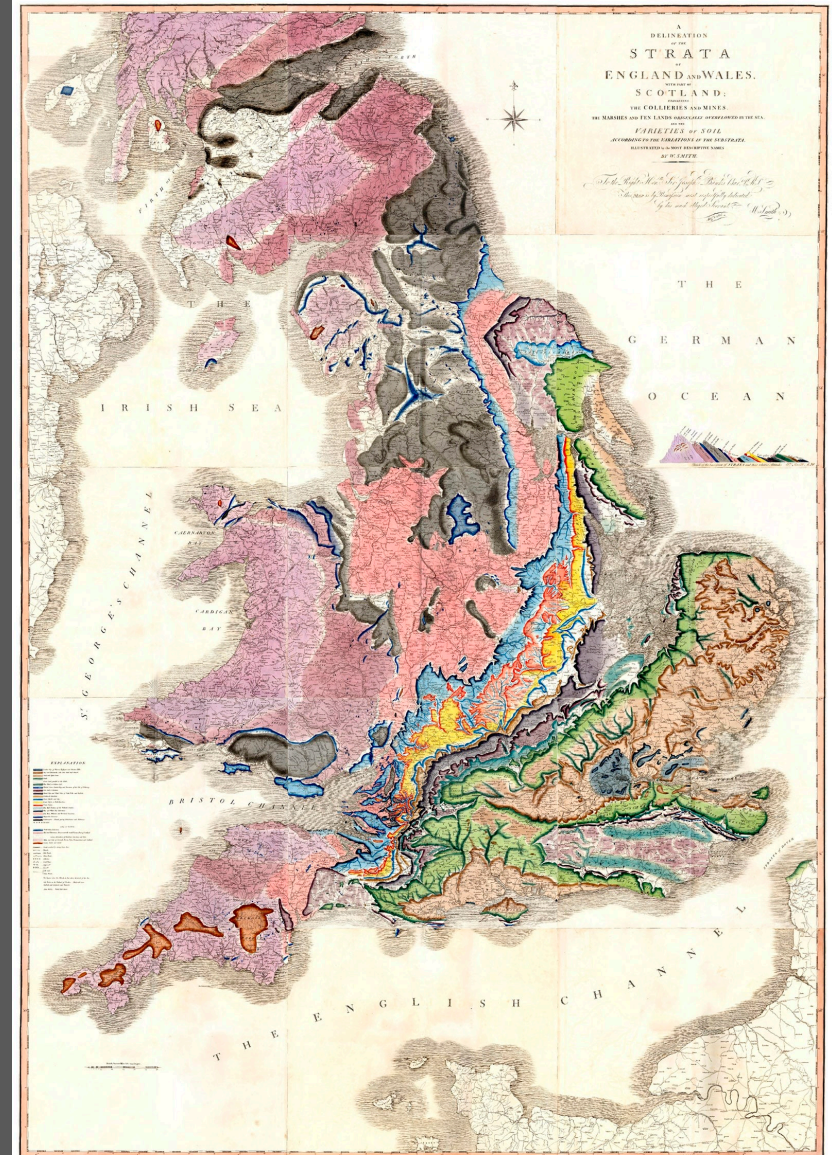
Pointer fra sidste forelæsning:

1. Nye astronomiske observationer og teorier: Jorden, en planet blandt andre planeter
2. Jorden gennemløber et livsforløb betinget af mekaniske naturlove
3. Nicolaus Steno: Jordskorpen foranderlig og kan studeres historisk gennem geologisk analyse (stratigrafi)

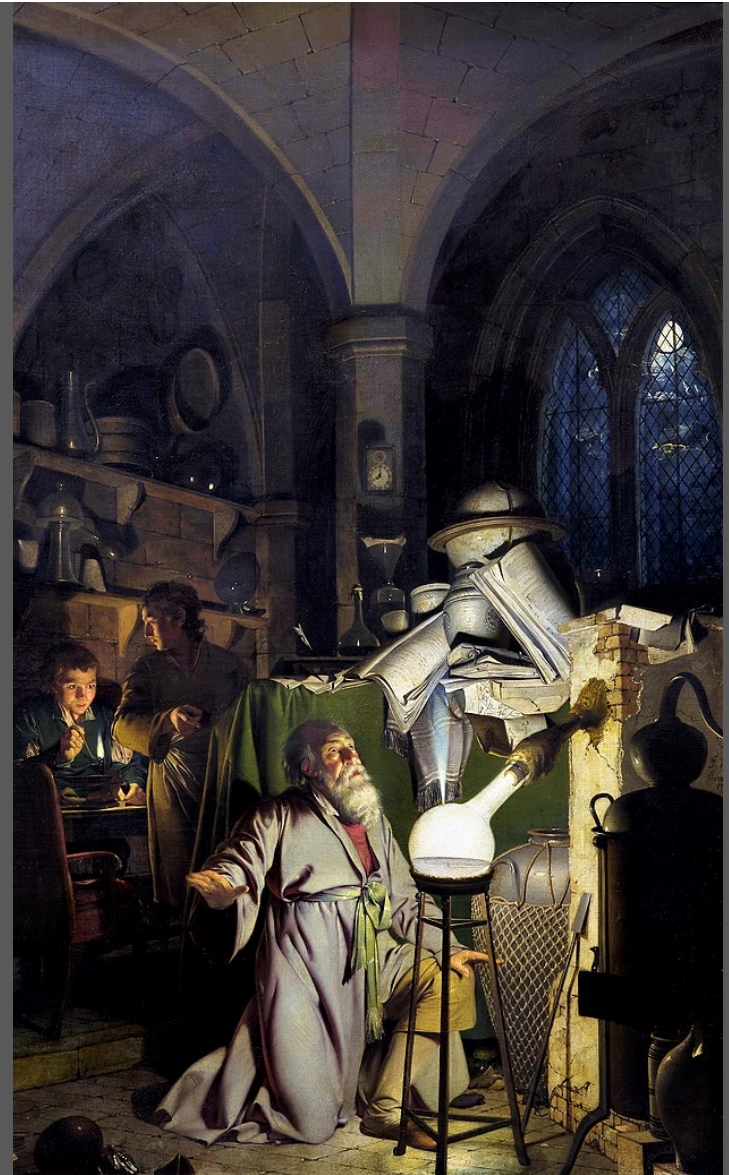


Hvor skal vi hen i dag?

1. Ny forestilling om Jordens perioder
2. Betydning af minedrift for geologien
3. Geologiens regelmæssighed: Uniformisme eller katastrofisme?
4. Jordens alder i 1900-tallet



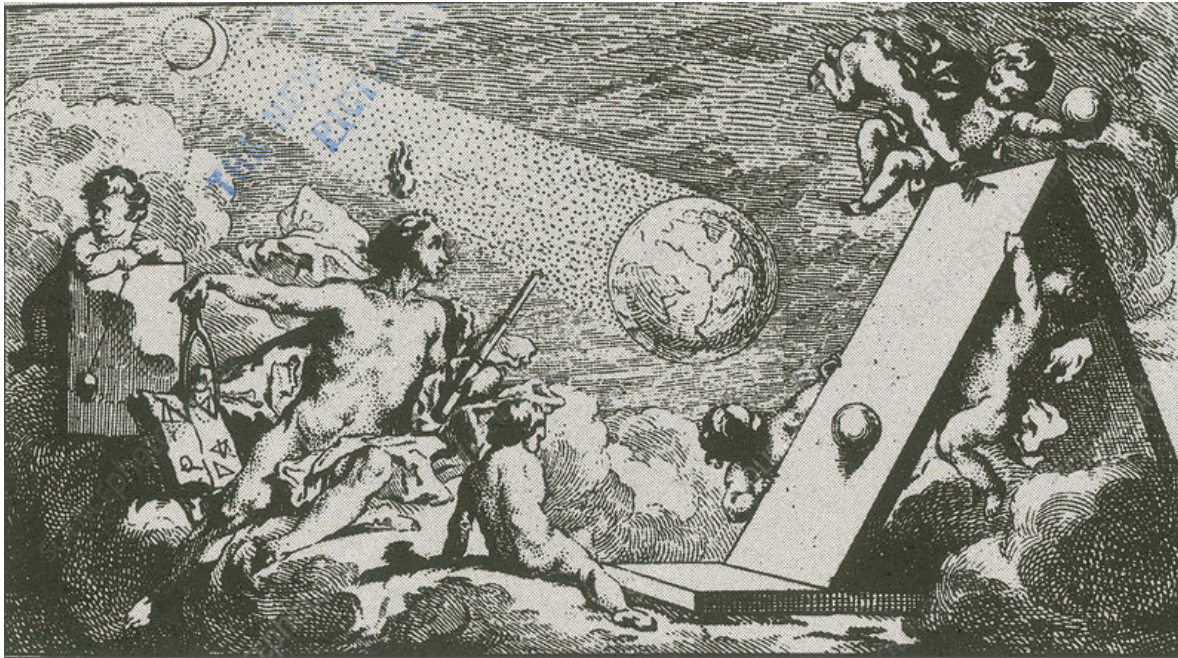
Jordens epoker





Georges Buffon (1707–1788)

1. Naturhistoriker, bestyrer af den kongelige botaniske have i Paris (Jardin du Roi)
2. Transformerer den botaniske have til et eksperimentelt laboratorium for botanik, mineralogi og zoologi
3. Indfører Newtons fysik i Frankrig (s. m. Voltaire)
4. Beskriver naturen som én stor kæde, hvor alt er levende, men i varierende grader
5. Udgiver *Histoire Naturelle* i 36 bind mellem 1749 og 1788



Buffons teori om Jorden

1. Buffon udgiver sin teori om Jorden i *Histoire Naturelle*
2. Descartes: Jorden en udslukt sol
Buffon: Jorden et stykke jern, som en komet slog af Solen
3. Jernet har derefter kølet ned
4. Eksperimenter i Montbard: Hvor hurtigt køler jern ned?
5. Buffons estimat: Jorden er 70.000 år gammel

HISTOIRE
NATURELLE,
GÉNÉRALE ET PARTICULIÈRE,
AVEC LA DESCRIPTION
DU CABINET DU ROI.

Tome Dixième.



A PARIS,
DE L'IMPRIMERIE ROYALE.

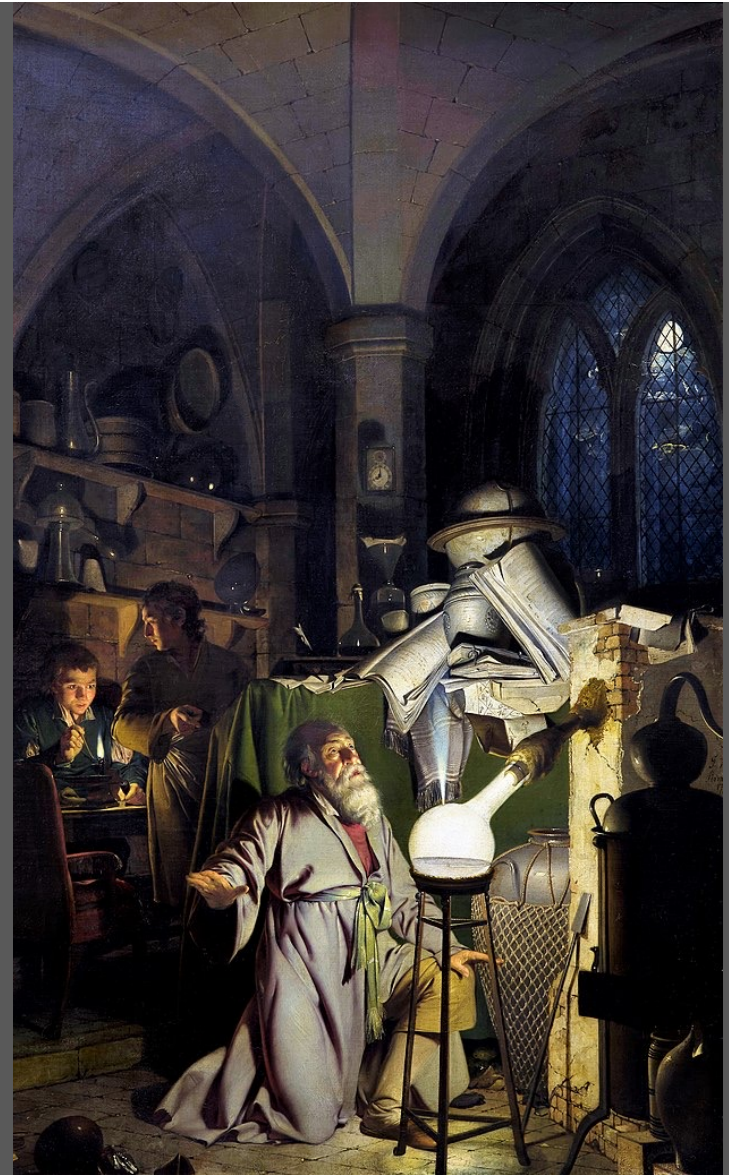
M. DCCCLXIII.



Buffons teori om Jorden

1. Stadieteori: Dyr og planter er økologiske væsner: De er afhængige af et miljø
2. Derfor har forskellige dyr og planter levet på forskellige tidspunkter (epoker)
3. Liv som afledt effekt (mennesket som afledt effekt!)
4. Jordens alder => Jordens aldre

Geologi og minedrift





Geologi og minedrift

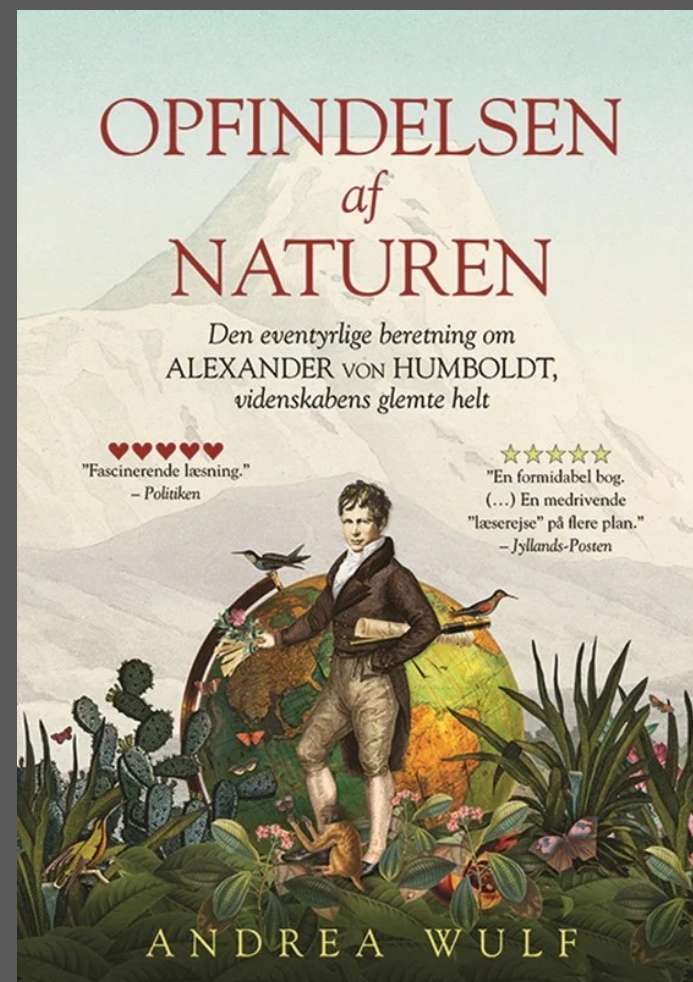
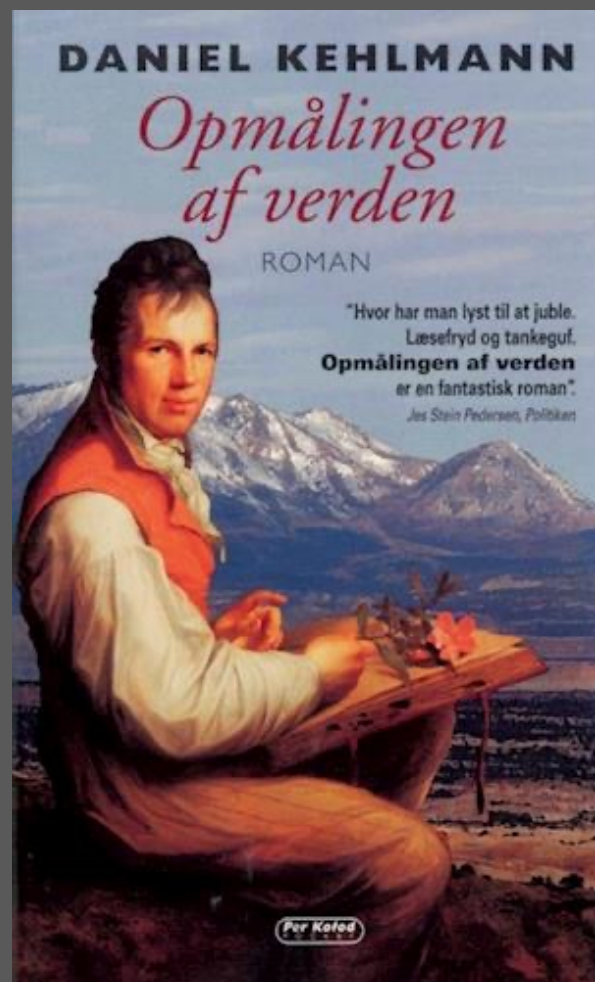
1. Videnskab bliver ikke (kun!) dyrket for sandhedens skyld: Økonomiske og politiske incitamenter
2. Hvorfor graver vi i jorden? For at skaffe ressourcer
3. Minedriftsmanualer som genre siden 1500-tallet
4. Abraham Gottlob Werner: Lærer i mineralogi på mineakademiet i Freiberg i slutningen af 1700-tallet
5. Stenarter aflejrer sig lagvist og på samme måde alle steder => metode til at identificere forekomster af vigtige metaller, mineraler og kul





Alexander von Humboldt

1. Studerede ved Werners mineakademi i Freiberg
2. Ekspeditioner til Sydamerika sammen med Aimé Bonpland
3. Plantegeografi, klimatiske zoner, vertikal kartografi



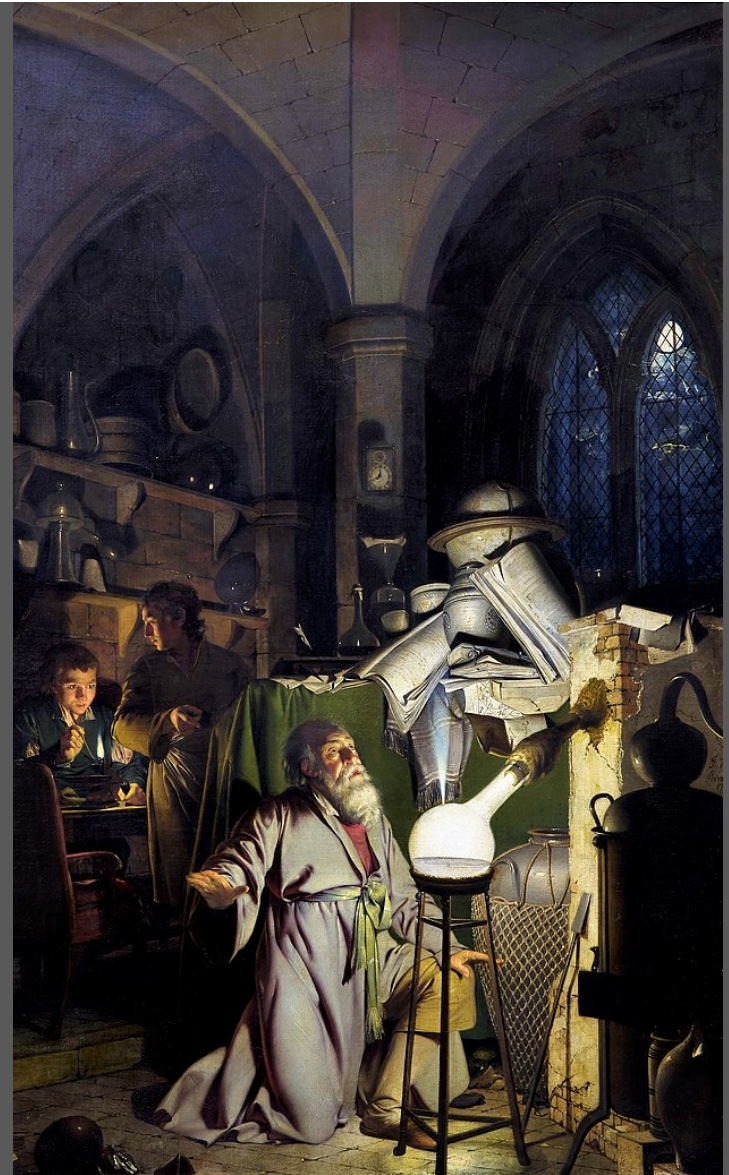
Geologi og minedrift: Fossiler

1. Werner og superpositionsprincippet: Samme klipper dannes på forskellige tidspunkter => Fokus på fossiler
2. Georges Cuvier: Der findes uddøde arter (fx mammutter)
3. 1840'erne: Palæozoikum, Mesozoikum, Kænozoikum
4. Mesozoikum: Reptilernes æra
5. Midten af 1800-tallet: Fascination af reptiler (=dinosaurer)





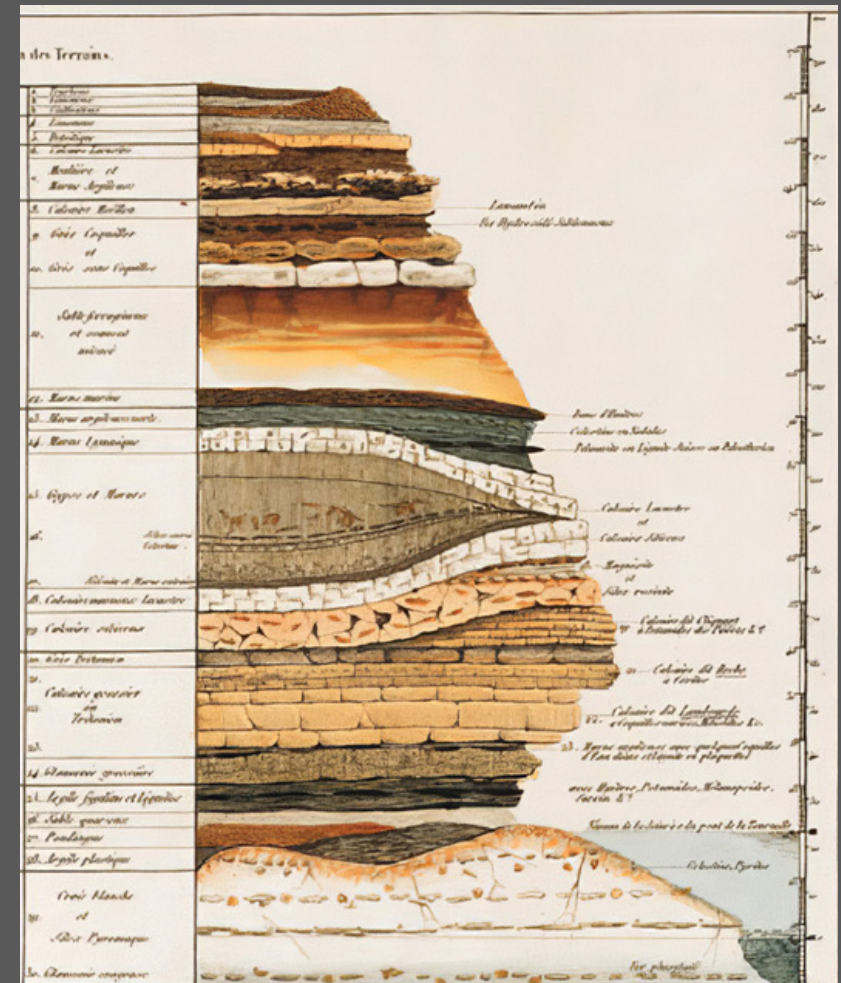
Katastrofisme og uniformisme





Katastrofisme

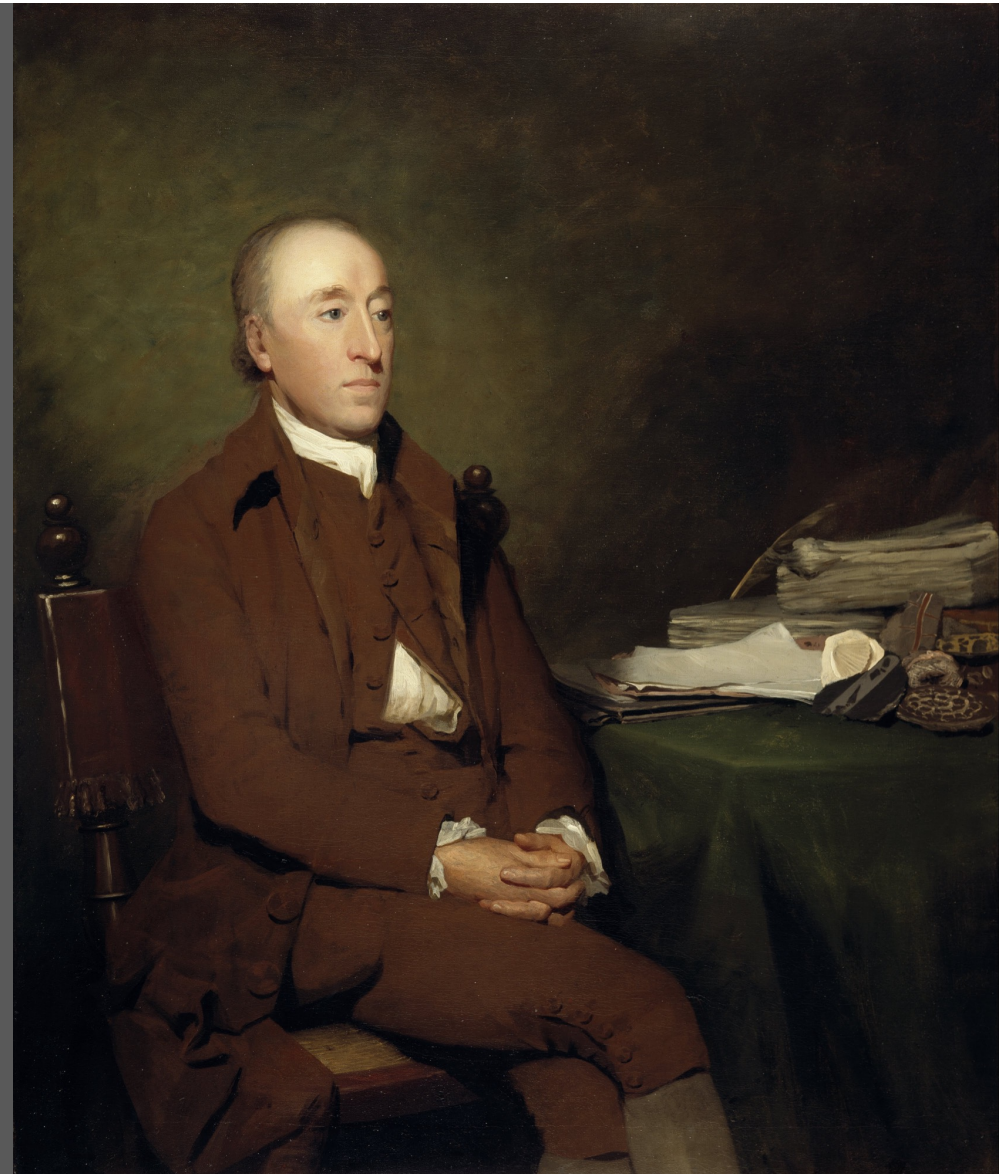
1. Georges Cuvier: Empiri fra fossiler
=> abrupte forandringer
2. Jordskælv, vulkanudbrud, tsunamier?
3. Jorden køler ned (jf. Buffon):
Voldsommere vulkaner og jordskælv i
forhistorisk tid
4. Jordens skorpe er fast => Pludselige
forandringer
5. William Buckland: Den bibelske
syndflod?



Uniformisme

1. James Hutton, *Theory of the Earth* (1795)
2. Kerneidé: De samme kræfter har virket til alle tider + vi kan ikke antage årsager, som vi ikke kan observere
3. Fokus: Erosion, langsomme forandringer
4. Ikke tsunami, men langsom påvirkning af vand
5. Cyklisk kosmologi: Jorden køler ikke ned, men forandrer sig blot (frem og tilbage)
6. Religiøst motiveret: Gud har sat skaberværket i gang, men intervenserer ikke (deisme)

James Hutton



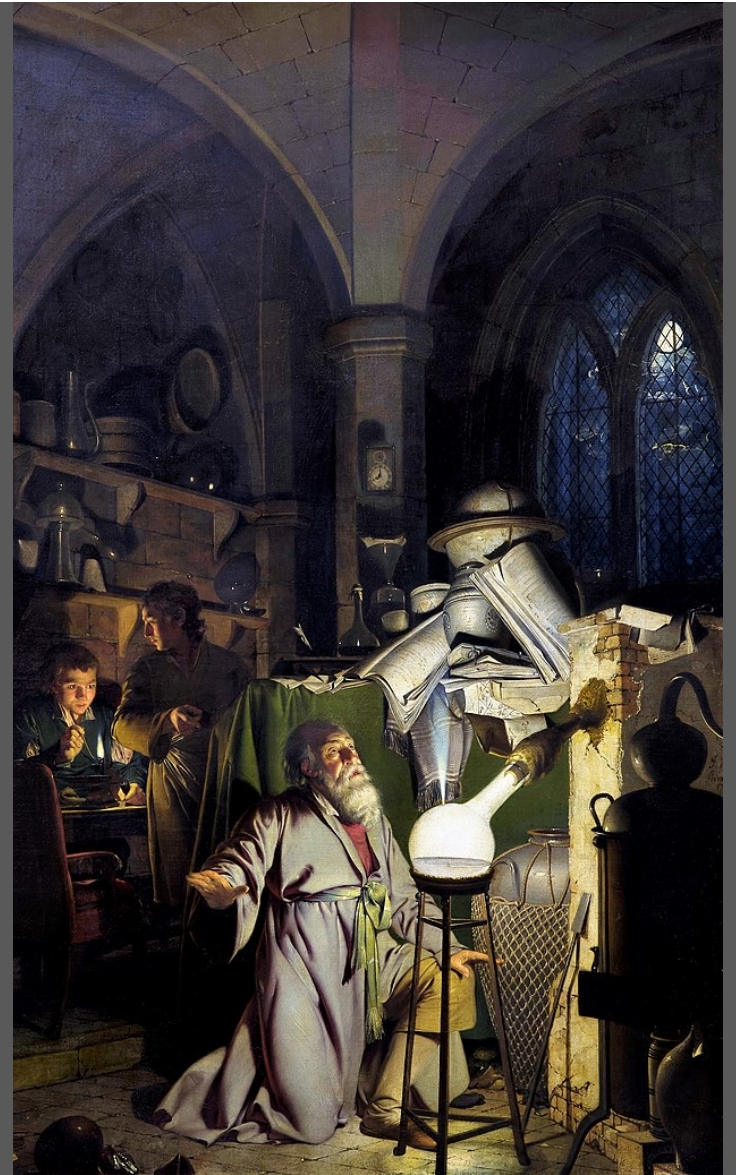
Uniformisme

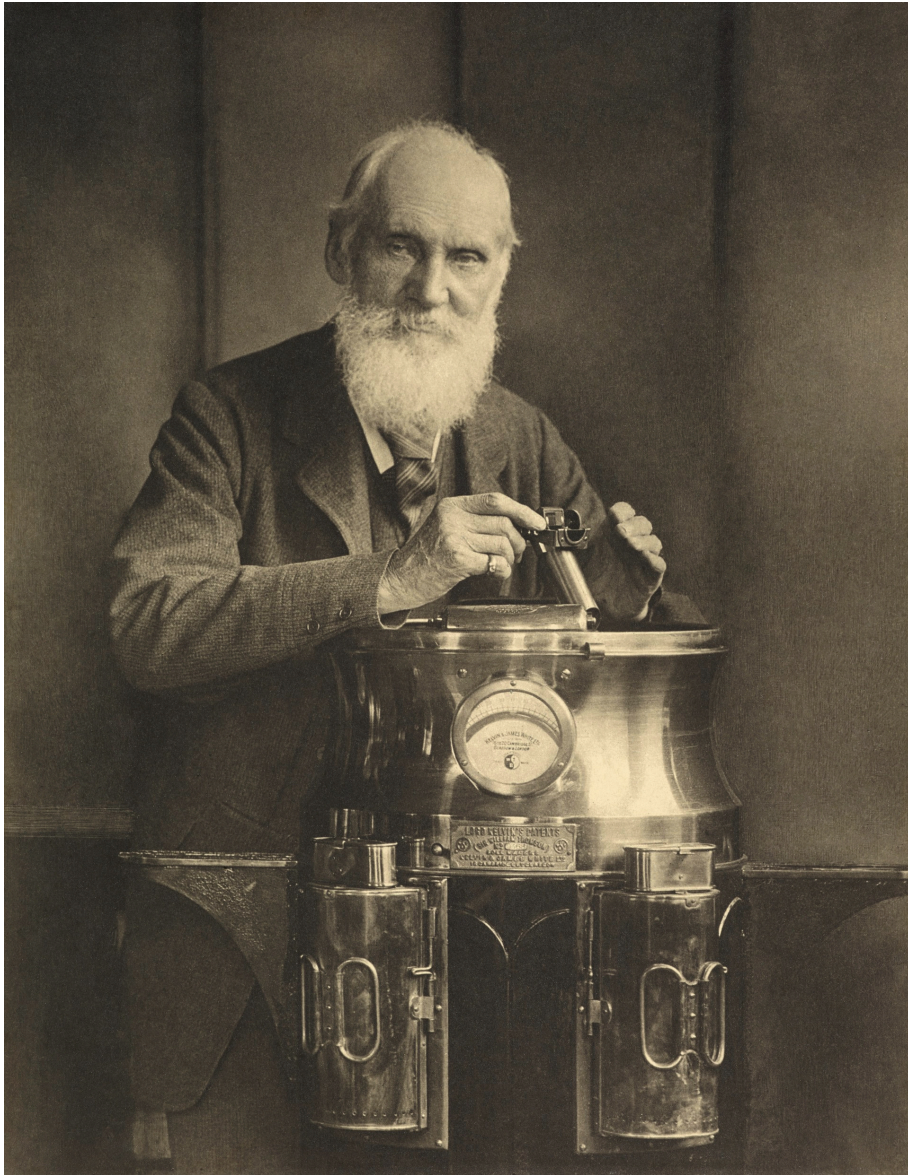
1. Charles Lyell, *Principles of Geology* (1830–33)
2. Empiriske studier af vulkaner, jordskælv og erosion, bl.a. vulkanen Etna i Sicilien
3. Pludselige hop i aflejringer: Skyldes at nogle tider ikke aflejres
4. Ahistorisk? Pattedyr fandtes også i tidligere epoker
5. Metodisk fremsynet (empirisk), men havde Hutton og Lyell ret i deres grundlæggende syn på Jordens historie?

Charles Lyell



Jordens alder i 1900-tallet





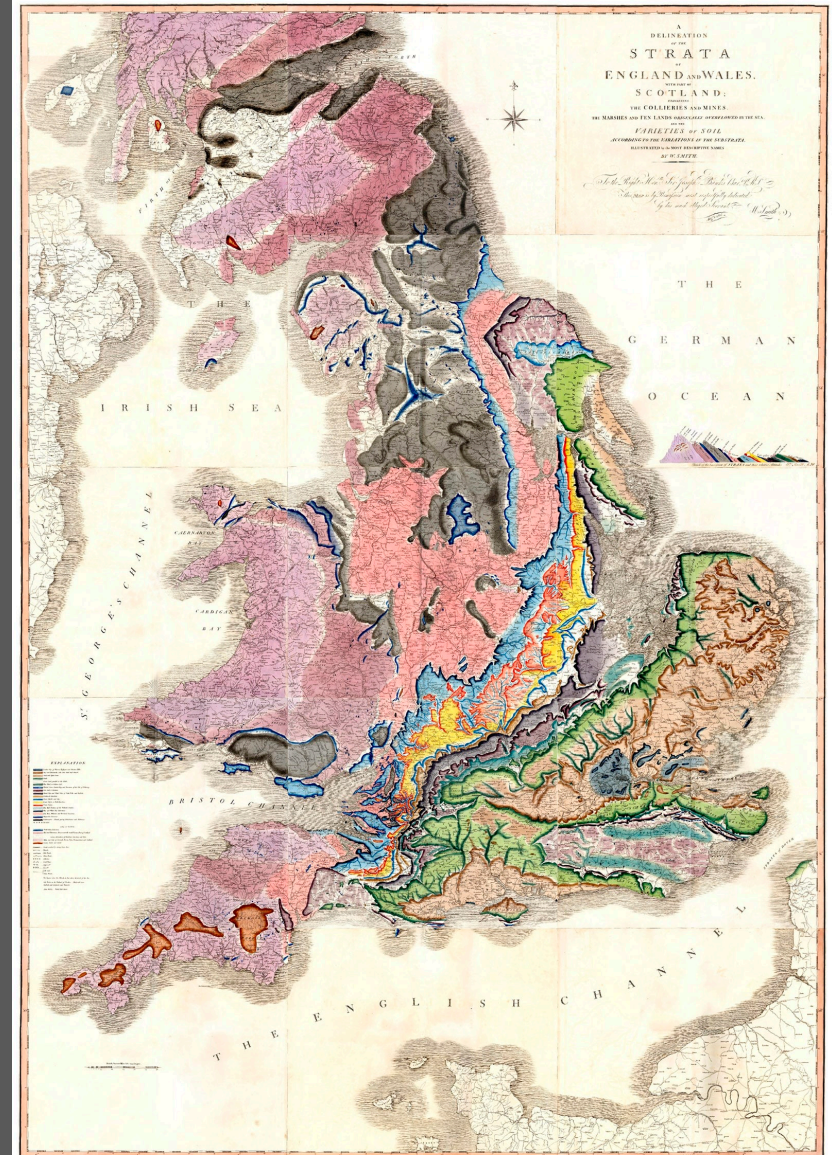
Jordens alder i 1900-tallet

1. Hvordan gik vi fra 70.000 år til millioner af år til 4,5 milliarder år?
2. William Thomson (Lord Kelvin):
 1. Termodynamik og entropi
 2. Jorden afkøles over tid (medmindre energi tilføres)
 3. Jorden må være ca. 25-200 mio. år gammel
3. Uniformistiske geologer: Det er ikke tid nok!
4. 1896: Opdagelsen af radioaktivitet => henfald danner energi/varme => Jordens temperatur er stabil (omtrent, ikke helt)
5. Grundstoffers halveringstider: Datering af Jorden til 4,5 mia. år

William Thomson, Lord Kelvin

Konklusioner

1. Jorden bliver et historisk fænomen, der kan studeres empirisk
2. Jordens alder udvides fra 4.000 år til 4,5 mia. år
3. Jordens historie inddeles i perioder
4. Livet på Jorden er foranderligt



Biblical Flood and geological deluge: the amicable dissociation of geology and Genesis

MARTIN J. S. RUDWICK

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Abstract: This paper summarizes debates, among European geologists in the early nineteenth century, about the possible equivalence (or non-equivalence) between the biblical account of Noah's Flood, and new and cumulative evidence for an exceptional watery catastrophe or 'geological deluge' in very recent Earth history. The 'diluvial theory' deserves to be taken seriously as an attempted explanation of some extremely puzzling physical features (many of them reinterpreted later as traces of a glacial 'catastrophe' or Ice Age). The 'geological deluge' was eventually recognized as having been far earlier in Earth history than any event recorded by literate human societies. Among geologists, although not always among the wider public, this gradual dissociation between biblical Flood and geological deluge was generally amicable, not acrimonious. It was facilitated by the concurrent development of biblical scholarship, which showed that earlier literalistic interpretations were no longer tenable (and were also destructive of religious meaning). What was transposed into geology in the course of these debates was the strong Judaeo-Christian sense that the world has had a directional and contingent history, which might have been punctuated by occasional catastrophic events.

Historical work on the relations between any of the natural sciences and any of the world's religions should always specify clearly the period, the place and the persons that are under discussion, and also define which social groups were involved. Historians of the sciences now rightly reject attempts to impose any single or simple pattern of, for example, either endemic conflict or perennial harmony, because these are incompatible with the variety and complexity of the historical evidence (see, e.g. Brooke 1991).

This paper is about the 'when' of the late eighteenth and early nineteenth centuries; the 'where' of the whole of Europe (including its offshore islands of Britain and Ireland); and the 'who' of the naturalists who at that time were beginning to call themselves 'geologists'. These are the parameters that Karl von Zittel (1839–1904) used when, over a century ago, he famously defined the decades around 1800 as 'das heroische Zeitalter der Geologie' (von Zittel 1899, p. 76). He too focused on research by Europeans, because during geology's 'heroic period' North America and the rest of the world were still marginal to high-level scientific debate (although treated as valuable sources of new observations and specimens). And he too concentrated on the social group of the leading scientific figures, and on what they were doing and discussing among themselves, rather than on the dissemination and reception of their ideas among the wider public (the latter is a very different story: for the British case, see O'Connor 2007b).

Within these parameters, this paper focuses on one celebrated (or notorious) case of the relation between geology and religion, namely the attempts to interpret certain physical features as traces of the Flood recorded in the book of Genesis, and conversely the attempts to reject any such correlation.

Earth's history, and its timescale

In late eighteenth- and early nineteenth-century Europe, during what political historians often call the Age of Revolution and the subsequent Age of Reform, the Earth sciences were radically transformed by becoming deeply historical in their outlook and practice. Those who pursued these sciences came to recognize not only that the Earth as a whole has had its own history, but also that its features (particular mountains, volcanoes, rocks, fossils and so on) likewise have specific histories built into them, and cannot be understood solely in terms of unchanging ahistorical 'laws of nature'. This radically new outlook on the natural world was the result of a deliberate transposition of methods and concepts from the writing of human history into the study of the Earth, to reconstruct the Earth's own history (in modern terms, geohistory) in all its complex particularity (Rudwick 2005, 2008).

This geohistorical research strategy was so successful and productive that most modern geologists are unaware of it. They use it routinely and take it

Teoretiske øvelser

- Religion og videnskab
- Syndfloden: Bibelsk eller geologisk fænomen?
- Fremkomsten af empirisk geologi

Næste uge: Brad om
videnskabelige revolutioner

